

Journal of the Air & Waste Management Association

ISSN: 1096-2247 (Print) 2162-2906 (Online) Journal homepage: www.tandfonline.com/journals/uawm20

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To cite this article: James J. Schwab, Henry D. Felton, Oliver V. Rattigan & Kenneth L. Demerjian (2006) New York State Urban and Rural Measurements of Continuous $PM_{2.5}$ Mass by FDMS, TEOM, and BAM, Journal of the Air & Waste Management Association, 56:4, 372-383, DOI: 10.1080/10473289.2006.10464523

To link to this article: https://doi.org/10.1080/10473289.2006.10464523

Published online: 29 Feb 2012.

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TECHNICAL PAPER

ISSN 1047-3289 *J. Air & Waste Manage. Assoc.* **56**:372–383
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New York State Urban and Rural Measurements of Continuous $PM_{2.5}$ Mass by FDMS, TEOM, and BAM

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ABSTRACT

Field evaluations and comparisons of continuous fine particulate matter ($PM_{2.5}$) mass measurement technologies at an urban and a rural site in New York state are performed. The continuous measurement technologies include the filter dynamics measurement system (FDMS) tapered element oscillating microbalance (TEOM) monitor, the stand-alone TEOM monitor (without the FDMS), and the beta attenuation monitor (BAM). These continuous measurement methods are also compared with 24-hr integrated filters collected and analyzed under the Federal Reference Method (FRM) protocol. The measurement sites are New York City (the borough of Queens) and Addison, a rural area of southwestern New York state. New York City data comparisons between the FDMS TEOM, BAM, and FRM are examined for bias and seasonality during a 2-yr period. Data comparisons for the FDMS TEOM and FRM from the Addison location are examined for the same 2-yr period. The BAM and FDMS measurements at Queens are highly correlated with each other and the FRM. The BAM and FDMS are very similar to each other in magnitude, and both are ~25% higher than the FRM filter measurements at this site. The FDMS at Addison measures ~9% more mass than the FRM. Mass reconstructions using the speciation trends network filter data are examined to provide insight as to the contribution of volatile species of $PM_{2.5}$ in the FDMS mass measurement and the fraction that is likely lost in the FRM mass measurement. The reconstructed mass at Queens is systematically lower than the FDMS by ~10%.

IMPLICATIONS

Fine particle pollution is a major health and regulatory concern in the United States and around the world. In the United States, the FRM for measuring $PM_{2.5}$ has been shown to suffer from measurement artifacts that are dependent on season and location. A second drawback of the FRM is that the data are not available in real-time or near real-time, which means the data are less useful for public notification of pollutant concentrations and for use in forecasting models. This paper presents results from continuous $PM_{2.5}$ measurement methods in New York state, and evaluates the methods in comparison to the FRM. Collocated filter measurements from the speciation trends network, along with a model-based retained water estimate, are used to explain or reconstruct $PM_{2.5}$ measurements from the FDMS TEOM monitor and the FRM. The ability to determine the seasonal contributions of the species that comprise the FRM measurement could be a great benefit to state regulators, health investigators, and others. $PM_{2.5}$ control strategies will inevitably target some species over others. The knowledge of which species make up the FRM measurement and their seasonal dependence will enable a more accurate determination of the potential effectiveness and possible health outcomes of any proposed regulation based on FRM measurements.

INTRODUCTION

As part of U.S. Environmental Protection Agency (EPA)–sponsored Particulate Matter Technology Assessment and Characterization Study–New York (PMTACS–NY) Supersite Program, the New York State Department of Environmental Conservation (NYSDEC) and the Atmospheric Sciences Research Center (ASRC) at the University at Albany have collaboratively operated urban and rural measurement sites with collocated filter dynamics measurement system (FDMS) tapered element oscillating microbalance (TEOM) monitors, standard TEOM monitors, beta attenuation monitors (BAMs), and Federal Reference Method (FRM) samplers for the past several years. Part of the research design, and a major program goal of the PMTACS-NY program, includes the evaluation of emerging instrumentation and technology transfer from the research arena to routine application. These new instruments are operated and evaluated by a mixture of university and state agency staff with the expectation that the final commercial version of each instrument will be reliable and

valuable to both research operations and to routine state agency air monitoring networks.

Continuous fine particulate matter ($PM_{2.5}$) mass monitoring is performed by research, state, local, and tribal organizations for a variety of purposes including (1) air quality forecasting, (2) near real-time public health reporting, (3) model and State Implementation Plan (SIP) evaluation, (4) trends analysis, (5) analysis of intense short-time scale exposures, and (6) as a replacement for FRM sampling. Continuous mass measuring instruments that were developed or improved in the past several years are currently being evaluated with the expectation that they will eventually be deployed in a measurement network capable of near real-time public reporting and forecasting. The 1-hr time resolution data also lend more readily to the source attribution and control evaluation studies required for SIP evaluation. Continuous methods must be thoroughly evaluated, compared with existing measurement technologies, and documented before they can be reliably deployed in a public reporting agency's monitoring network.

Integrated 24-hr filter samples collected using a FRM sampler form the basis for the National Ambient Air Quality Standards (NAAQS) for $PM_{2.5}$. There are known and reported artifacts associated with this method.[1–5] $PM_{2.5}$ is a complex mixture of chemical species in a range of physical sizes that varies significantly with season and location. The FRM may be adequate in most instances to determine compliance or noncompliance with the NAAQS. We know, however, that the FRM does not measure "true atmospheric $PM_{2.5}$ mass concentration," and the biases of the FRM, like the particulate matter itself, vary with location and season.[1,6,7] The FRM most often undermeasures the true $PM_{2.5}$ mass, as shown in many studies,[2,3] so this persistent negative bias sets up a conflict or a difference as to how we understand, measure, and report concentrations of $PM_{2.5}$. The entities being regulated (states, tribes, and localities) might reasonably prefer to continue using the FRM for regulatory purposes because the data are more consistent with that used to develop the NAAQS. However, the new measurement techniques report systematically higher mass concentrations than measured by the FRM, which may in the future require some reinterpretation of $PM_{2.5}$ NAAQS, because the health-based epidemiological time series are based on FRM $PM_{2.5}$ mass data. Most importantly, verifying that these advanced measurement techniques more closely represent the true $PM_{2.5}$ mass in the atmosphere and understanding the source of these systematically different results has significant implications in the times-series analyses of health outcomes. In particular, health effects associated with systematic seasonal differences in reported FRM mass may be masked and misinterpreted in epidemiological time-series analyses in consideration of the seasonal effects of confounding meteorological variables on health outcomes. An additional but somewhat speculative outcome might be that the species composition of lost FRM mass is potentially more potent in its toxicology, confounding the $PM_{2.5}$ mass–health outcome relationship.

This paper reports on the deployment, use, and evaluation of recently developed or improved methods for the continuous measurement of $PM_{2.5}$ mass concentrations. This is a topic of considerable current interest, as noted above, and has been addressed by a number of the PM supersites programs.[8,9] Special attention is devoted to the FDMS TEOM developed by Rupprecht and Pataschnick, Inc. (Albany, NY) because this instrument reports values that can be interpreted as the volatile and nonvolatile contributions to the measured $PM_{2.5}$ mass. The other semicontinuous measurement method (i.e., producing hour averaged measurement values) showing promising results is the beta attentuation method.[10–13] Data from a BAM deployed in Queens are compared with data from the FRM and a collocated FDMS. Two sites in New York State are involved in this study; the borough of Queens in New York City and Addison, in the rural Southern Tier of New York state. New York City data comparisons between the FDMS TEOM, BAM, and FRM are examined for seasonality during a 2-yr period from January 2003 to December 2004. Comparisons between the FDMS TEOM and the FRM at Addison are examined for the same 2-yr period.

In addition to correlation plots comparing the various datasets, we examine the smoothed seasonality of mass concentration and chemical species concentrations using second-order polynomial fits. These fitted curves are used to reconstruct $PM_{2.5}$ mass from the species measured by the Speciation Trends Network (STN). Water retained by the aerosol is calculated from an empirical relationship based on a thermodynamic model.[14] These reconstructions help elucidate the volatile versus nonvolatile composition of the aerosol, and assist in the interpretation of the FRM and STN filter measurement data and that from continuous PM mass monitors like the FDMS and BAM. This is the type of information that can be used by state agencies who must understand the distribution and seasonality of the species or components of $PM_{2.5}$ as measured by the FRM. This knowledge can be used to predict how $PM_{2.5}$ control strategies that are targeted at specific species may result in lower FRM measurements in some seasons and unchanged FRM measurements in other seasons.

SITE INFORMATION

Basic site information is summarized in Table 1. The Queens and Addison sites were used in the earlier study of standard TEOM and Nafion dryer equipped TEOM monitors.[7] Brief additional information about the sites is given in the text below.

Table 1. Site information.

Site Name	Pinnacle State Park	Queens
Latitude	42.09°N	40.74°N
Longitude	77.21°W	73.82°W
Elevation above MSL	505 m	25 m
Structure	Trailer	2-story school
Classification	Rural	Urban
Land use	Forest/golf course	Residential/commercial
Start date, FDMS	3/2003	1/2003
Data capture, FDMS	57%	65%
Start date, BAM	8/2004	11/2003
Data capture, BAM	95%	99%
FRM Schedule	Daily	Daily

Queens, New York City

New York City Public School 219 (PS 219) is located between the Flushing and Forest Hills neighborhoods in the borough of Queens. The school is directly adjacent to the southeast corner of the Queens College City University of New York campus. The site is ~750 m south of the Long Island Expressway and ~1 km east of the Van Wyck Expressway, two of the busiest highways in New York City. The site is adjoined to the east by a ~1 km^2 cemetery, and Flushing Meadow Corona Park and Kissena Park (both large urban parks) are each less than 2 km from the site.

Pinnacle State Park

Pinnacle State Park in Addison, NY, is a rural site operated by ASRC. A very common synoptic flow pattern transports air from the industrial Midwest, over the Addison area, and eastward to the Eastern seaboard. The air monitoring trailer is located on the eastern side of Orr Hill, part of the Allegheny Plateau of southern New York state and northern Pennsylvania. The surrounding areas include a 9-hole golf course, a picnic area, undeveloped state forest lands, abandoned pastures, a 20 hectare pond, and a handful of homes. The village of Addison (population ~1800) is ~4 km to the northwest and the town of Corning (population ~12,000) is 15 km to the northeast.

MEASUREMENT METHODS

FDMS

The FDMS TEOM is a switched flow, self-referencing, differential measuring method based on the continuously measuring TEOM (Rupprecht & Patashnick, Albany, NY). The instrument is described by Grover et al.[15] The basic differential measurement approach is presented by Patashnick et al.[16] Descriptions and applications of the differential TEOM system using an electrostatic precipitator (ESP) to remove particles from the sample flow stream are presented in a number of recent papers.[8,17–19] The FDMS TEOM is similar in principle to the ESP differential TEOM, with the major hardware changes being the addition of a switching valve and the replacement of the ESP module with a chilled Pallflex filter for the creation of the particle-free sample flow stream. The FDMS has been designated as an equivalent measurement method for $PM_{2.5}$ by the state of California.[20]

The flow system consists of an inlet with a coarse particulate matter (PM_{10}) separator, followed by a sharp-cut $PM_{2.5}$ cyclone.[21] The total flow of 16.7 L/min is split into two parts, a sample flow and a bypass flow. The instrument sample flow of 3 L/min passes through a Nafion dryer bundle just upstream of a large-orifice switching valve. During the base mass period, the sample flow enters the TEOM sensor directly and the microbalance detects mass changes on the molded filter capsule caused by particles in the airstream and artifact effects such as evaporation, condensation, and adsorption. During the reference mass period, the sample flow is directed through a 47-mm diameter Pallflex Emfab TX40 HL20 filter membrane enclosed in a temperature-stabilized housing. The current design maintains the 47-mm filter and filter holder at a temperature of 4 °C. During the reference mass period the particles are captured onto the 47-mm filter and effectively retained because of the cold temperature, whereas the ambient air flows back into the sample tube and through the TEOM filter capsule. During this reference period the microbalance detects mass changes on the filter capsule due only to artifact effects. The derived mass concentration readings from the FDMS are the 1-hr running averages resulting from computation of the difference between the running averages of the measured values during the base and reference periods.

One disadvantage of the FDMS system is that very fast (on the order of minutes) changes in ambient concentrations are not tracked accurately, because the valve switching time is set to six minutes to attain high signal-to-noise and low detection limits. A major advantage of the FDMS system, and a feature not available with any other instrument to our knowledge, is that the base and reference mass measurements are both available as output values, as is the derived difference—the reported mass concentration. The reference mass concentration in particular can, in many cases, be dominated by the evaporation of volatile particles, making this value a good estimate of the volatile component of the $PM_{2.5}$. Likewise, given the same conditions, the base mass is a good estimate of the nonvolatile component of the $PM_{2.5}$ mass. Note that this situation only holds when condensation and adsorption effects (and instrument noise) are much smaller than the mass change from evaporation. We will make use of this estimate of the volatile component of the PM below.

BAM

Beta absorption was first developed as a method for the measurement of airborne PM in the 1960s and 1970s.[10,11] Since that time there have been a number of research and commercial instruments based on this method. A number of recent papers have described the hardware and operation of the beta gauge[13] and presented evaluations and data comparisons for various versions of this instrument.[12,20,22,23] A number of BAMs have been designated as automated equivalent methods for the measurement of PM_{10} by EPA;[24] California has designated the BAM as an equivalent method for $PM_{2.5}$[20]. The BAM used in this work was Model 1020 produced by Met One Instruments (Grants Pass, OR). Like the FDMS (and the FRM), the BAM uses an inlet flow of 16.7 L/min. The size selective inlets are identical to the FDMS: a PM_{10} inlet followed by a $PM_{2.5}$ sharp-cut cyclone.[21] The full 16.7-L/min inlet flow is used as the sample flow for this instrument. Condensation is controlled on the filter material by utilizing an optional "Smart" heater that cycles on and off to automatically maintain the sample relative humidity (RH) below 45%.

TEOM

The TEOM has been described in the literature by Patashnick and Rupprecht.[25] The instrument continuously draws a sample airflow over a filter capsule mounted onto the end of a hollow quartz tube (the tapered element). This tube undergoes a forced harmonic oscillation, and the frequency of this oscillation is tracked to provide a direct measure of the mass changes of the system due to the particles collected onto the filter. There are both positive and negative artifacts associated with this method,

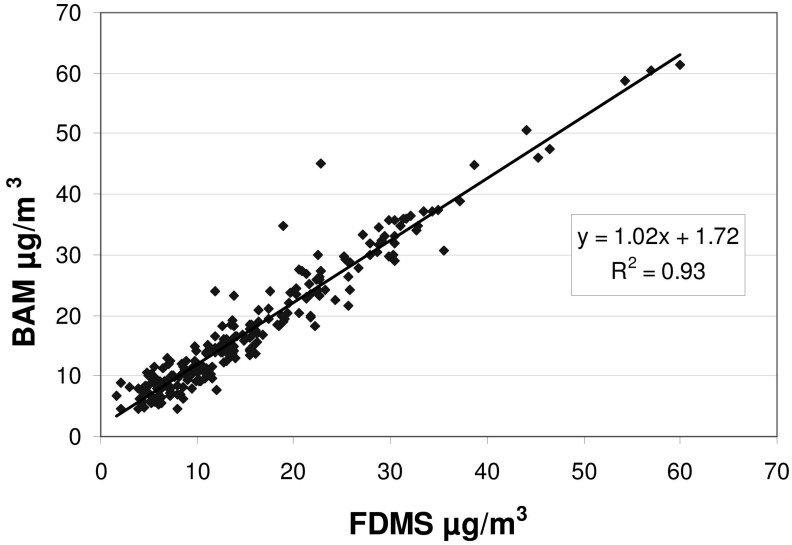


Figure 1. Correlation scatter plot and fitted linear regression line for the 2004 24-hr averaged mass concentration data from the FDMS TEOM and BAM monitors in Queens. Also shown is the equation of the fitted regression line and coefficient of determination or R^2 value.

and a number of papers have addressed these issues and the issues of comparability of this method with the FRM.[2,7,8,22,23] The TEOM monitor has been designated an "automated equivalent method" for the measurement of PM_{10} by EPA.[24]

FRM

The FRM sampler has been defined, along with the Ambient Air Quality Surveillance Program and the Quality Assurance Program in a series of Federal Register publications.[26–28] Papers by Noble et al.[29] and Peters et al.[30] give good descriptions of the method and the field performance of various FRM samplers. The FRM sampler used for this study was the Rupprecht & Patashnick Model 2025 Partisol-Plus sequential air sampler. The flow system consists of an inlet with a PM_{10} separator, followed by a well impactor ninety-six (WINS) $PM_{2.5}$ separator, collection on a filter, a flow sensor, and a pump. Flow (16.7 L/min), temperature, and atmospheric pressure sensors associated with the sampler are audited by the site operators monthly and by qualified personnel quarterly. Each of the sites in this study collects FRM filters on a daily basis.

STN

Integrated filter samples for chemical speciation are collected at the Queens and Addison sites according to the STN protocol. Rupprecht and Patashnick Model 2300 samplers are used to collect samples following the 1-in-3 days schedule distributed by EPA and shipped to Research Triangle Institute, NC, for analysis. The sampler has been described by Solomon et al.[31] In the STN protocol installations, the instrument consists of three sampling modules (Chem-Comb cartridges) connected to individual flow controllers and a single vacuum pump. The inlets are impactors with $PM_{2.5}$ cut points described by Demokritou et al.[32] Channel 1 contains a Teflon filter that is analyzed gravimetrically for mass and for elements by X-ray flourescence (XRF). The flow through this channel is 16.7 L/min. Channel 2 contains a single quartz filter, without denuder, which is analyzed for carbon (elemental carbon [EC], organic carbon [OC], total carbon [TC], and other empirical quantities) using the modified National Institute for Occupational Safety and Health 5040 method adopted by the EPA.[33,34] The flow through this channel is 10 L/min. Channel 3 includes a Na_2CO_3-coated honeycomb denuder to remove acid gases followed by a nylon filter, which is analyzed for nitrate, sulfate, ammonium, sodium, and potassium using ion chromatography. The flow through this channel is 10 L/min.

RESULTS

Data Validation

The FRM data were verified and flagged by the weighing laboratory and by NYSDEC. A few outliers were also excluded based on comparisons with collocated measurements and nearby measurements. FDMS and BAM measurements were collected and saved as hour-averaged data. Data from Queens were verified and flagged by NYSDEC, and data from Addison were verified and flagged by ASRC. Hour-averaged data were further averaged to 24-hr averages with the requirement that there be 21 valid hour averages to produce a valid 24-hr average. As can be seen from Table 1, a large amount of FDMS data is missing from both sites because of operational problems or the inability to validate the data. We are continuing to work with the manufacturer to increase the capture of valid data for the FDMS at our sites. Greater data capture by the FDMS would clearly improve the robustness of the results.

Comparisons of the Methods

Figure 1 shows the scatter plot of the 24-hr averaged FDMS TEOM and BAM data at Queens in 2004. There were 244 days with valid 24-hr averages for both instruments. The standard error for the slope coefficient is 0.02 and for the intercept coefficient is 0.33. These methods agree very well for this site, as evidenced by the high R^2 value and the slope very close to 1. The y-intercept of 1.72 $\mu g/m^3$ may indicate a systematic offset between the methods, but this comparison alone cannot determine if the

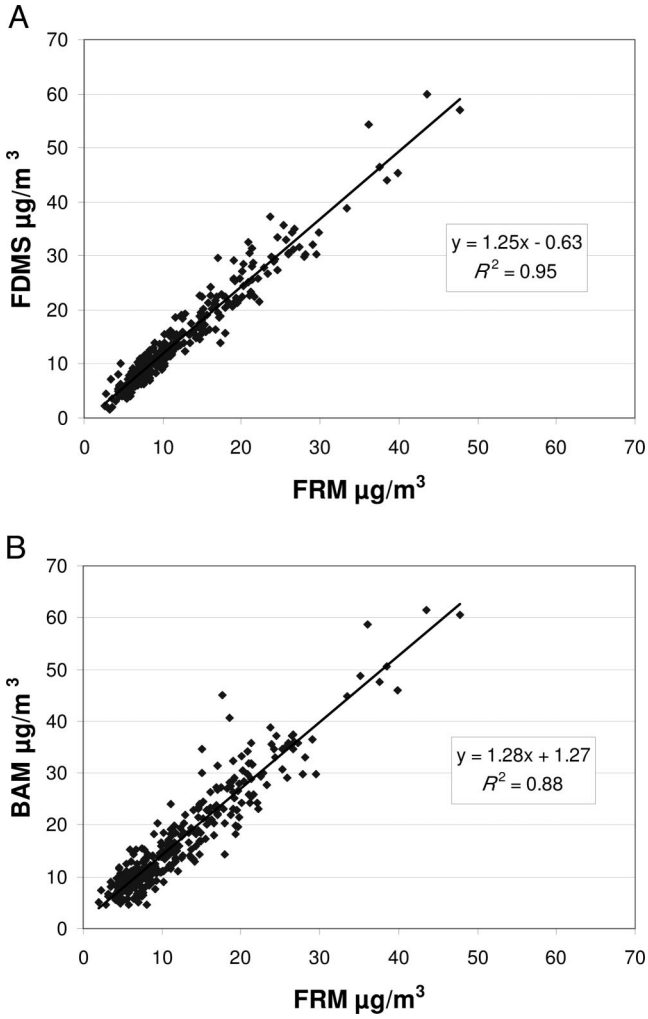


Figure 2. (a) Correlation plot for the 2004 24-hr averaged FDMS TEOM measurements vs the FRM measurements at Queens. The fitted linear regression line and coefficients are also shown. (b) Correlation plot for the 2004 24-hr averaged BAM measurements vs the FRM measurements at Queens. The fitted linear regression line and coefficients are also shown.

BAM is biased high or the FDMS is biased low, or a combination of the two.

Figures 2a and 2b show the 24-hr averaged FDMS and BAM measurement values plotted versus the FRM measurement values. Also shown are the least squares linear regression lines, the equation for the fitted line, and the R^2 or coefficient of determination. The standard errors for the slope coefficients are 0.02 and 0.03 and for the intercept coefficients are 0.26 and 0.38, respectively, for Figures 2a and 2b. There were 283 overlapping datapoints used in Figure 2a, and 320 datapoints used in Figure 2b. The regression slopes indicate that the FRM is missing ~25% of the particle mass on an annual basis at the Queens site in New York City. This result is similar to those seen recently is much shorter field studies in Utah and southern California.[15] The R^2 is higher for the FDMS versus FRM regression fit, and the y-intercept is closer to zero. Although not definitive, Figures 1, 2a, and 2b together hint that the BAM may have a small systematic offset of 1–2 $\mu g/m^3$ at this site. Inspection of Figures 2a and 2b also shows a number of points above and below the regression lines in each figure; in many of these instances it appears the FDMS and BAM agree with each other better than with the FRM. This could indicate either collection errors in the FRM operation at some or all of those dates or sampling intervals in which the $PM_{2.5}$ has a high percentage of volatile components that are undermeasured by the FRM.

The valid 2003 and 2004 FDMS data from Addison is plotted versus the FRM values in Figure 3. There are 349 datapoints from a 22-month period in 2003 and 2004 that are included in this plot. The standard errors for these slope and intercept coefficients are 0.02 and 0.18, respectively. Once again the correlation is quite high, and here we have an intercept of nearly zero. The intercept of nearly zero is different from the standard and Nafion dryer equipped TEOM monitors, which yielded significant negative intercepts when correlated with FRM data in an earlier paper.[7] The slope is much lower than that derived from the Queens data, indicating less volatility of the aerosol at this rural site. Still, it is clear that on an annual basis, the FDMS TEOM monitor captures nearly 10% more mass than the FRM at this site. Year-long comparisons between the FDMS and FRM measurements at three locations in Vermont are similar to the Addison results, showing the FDMS, on average, 4%, 7%, and 9% greater than the FRM.[35] The BAM at this site used Teflon sample tape instead of the usual quartz tape. However, the instrument was not modified to operate with this thinner tape, which led to unacceptable levels of leakage and unreliable data. For this reason BAM data from Addison are not included in this analysis.

Seasonal Dependence of Aerosol Mass and Components

The most dramatic swings in $PM_{2.5}$ mass concentrations are attributed to meteorological factors producing pollution events or very clean air masses. Very often a 4–7-day synoptic pattern in the PM mass concentration is evident at a given site. Sometimes the buildup and cleanout occur on shorter or longer time scales. Buried within the synoptic and other meteorological noise is a seasonal dependence of the average $PM_{2.5}$ mass concentrations and the major components of $PM_{2.5}$ mass. The seasonal pattern at these sites shows higher values of $PM_{2.5}$ mass, sulfate, and other components during the summer months at both of these sites.[36] The peak in average concentrations near the middle of the year suggests a simple, yet plausible, second order polynomial fit to describe the seasonal dependence. In using such a simplified fit, all information about short-term fluctuations is lost, so an analysis based on this methodology is aimed more at understanding average concentrations as opposed to peak concentrations. The technique is illustrated in Figure 4, where the 2004 FRM mass concentrations at Queens are plotted as a function of Julian day. A box plot of monthly averaged values would give a very similar result. Also shown is the curve representing the second order polynomial fit to this data and the equation of the curve. The failure of the curve to match up at the beginning and the end of the year is indicative of the inter-annual variability typically found

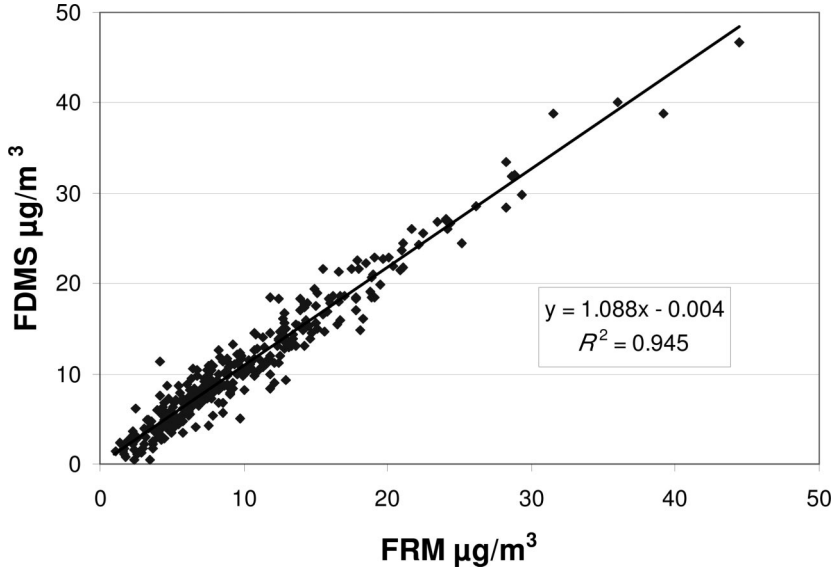


Figure 3. Correlation plot for 2003 and 2004 24-hr averaged FDMS TEOM measurements vs FRM measurements at Addison. The fitted linear regression line and coefficients are also shown.

in mass concentration data. Julian day plots and polynomial fits were also created for the FDMS and standard TEOM 2004 datasets at Queens. All fits showed the concentrations peaking in the middle of the year, with the FDMS and standard TEOM curves exhibiting greater curvature than the FRM curve. In addition to fitting these mass concentration datasets, we also plotted and fitted the differences between the FDMS and FRM values and between the FRM and standard TEOM values. These difference curves will be used below.

In an attempt to understand the seasonal behavior of the major components of $PM_{2.5}$ we have plotted and fitted these components from STN measurements versus Julian day, just as we did with the FRM, FDMS, and TEOM data. Since the STN data is only every third day, we used data from all of 2003 and 2004 for this analysis. Two examples of species with clear seasonal patterns are sulfate and nitrate, shown in Figures 5a and 5b, respectively. Sulfate peaks quite prominently in the summer, and the fitted curve indicates a 3–4 $\mu g/m^3$ difference between the mid-winter and mid-summer average values. Nitrate shows exactly the opposite behavior, peaking in the winter and reaching a minimum in the late summer. Other species fitted included ammonium, OC, EC, and a sum of the trace elements measured by XRF (excluding sulfur). Trace element masses are used as reported and not adjusted for oxygen mass in their typical oxides. There was little average seasonal variation in the trace elements, so an average value for the whole year is used. EC has a slight seasonal dependence at Queens, varying about ± 0.1 $\mu g/m^3$ above and below the average.

Particle bound water is estimated using an approximation based on the aerosol inorganic model.[37] The model uses the measured sulfate, nitrate, and ammonium on the filter, together with the equilibrium measurement temperature and RH values to determine the mass of particle bound water. In a paper by Frank,[14] the particle bound water is shown to have a high linear correlation

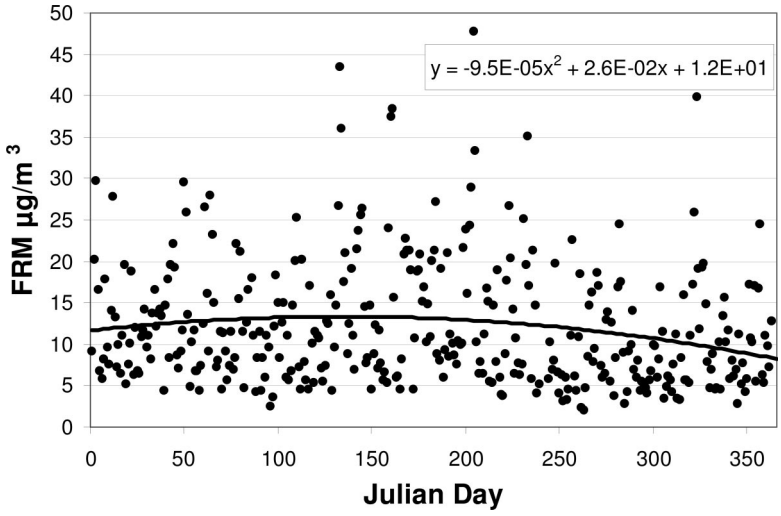


Figure 4. 2004 FRM measurements at Queens plotted as a function of Julian Day and fitted with a second order polynomial.

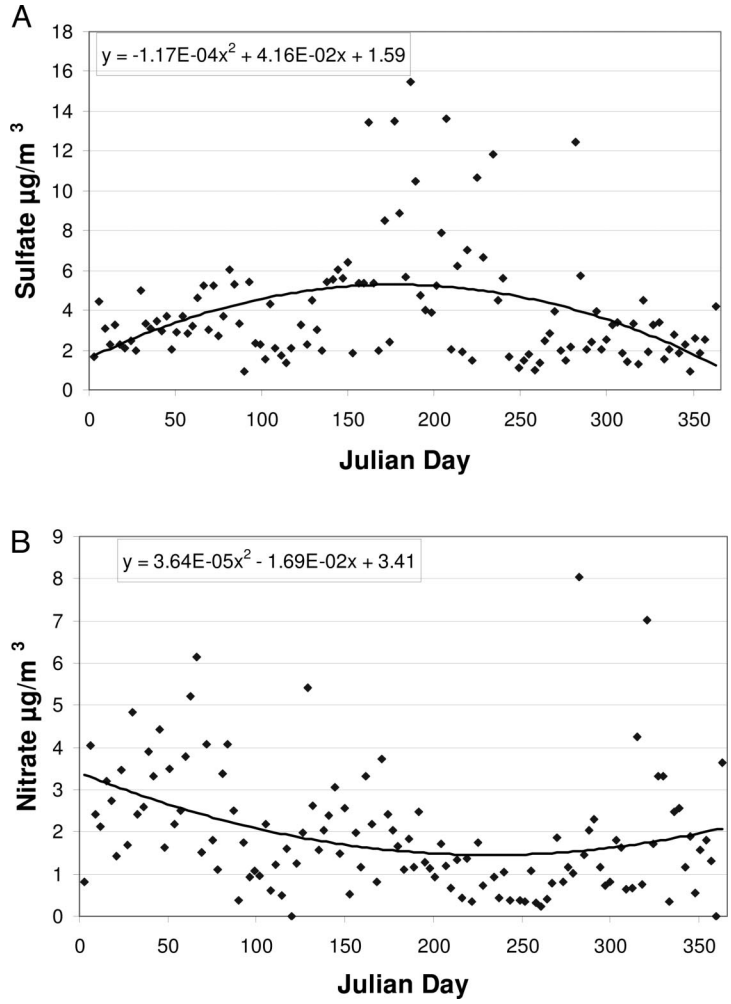


Figure 5. (a) Sulfate measurements from STN filters at Queens from 2003 and 2004 plotted as a function of Julian Day and fitted with a second order polynomial. (b) Same as (a), but for nitrate measurements from STN filters.

with the sum of sulfate plus total ammonium ion concentrations. At six sites in the eastern United States, the ratio of average $PM_{2.5}$ bound water to sulfate plus total ammonium concentrations was determined to be 0.24. We use this value to estimate particle bound water from the fits for sulfate and ammonium.

Figure 6 illustrates the result when the fitted species and estimated water are plotted along with the fitted FDMS and BAM curves. The STN species included are sulfate, nitrate, ammonium, OC, blank corrected and multiplied[38] by 1.6, EC, a constant value for the trace elements, and estimated water. The 1.6 OC multiplier used here is likely conservative, and its value is also seasonally dependent; refinement of the multiplier is under study and will likely be considered in future applications, but is currently beyond the scope of this study. The difference between the reconstructed mass from the STN species and the FDMS mass varies from ~1.4 $\mu g/m^3$ at the beginning and end of the year, to ~1.1 $\mu g/m^3$ in the summer, and is less than 10% on average. Increasing the OC multiplier from 1.6 to 2 nearly erases the difference in the summer and roughly halves the difference during the middle of the winter. Another possible explanation of the difference is that the FDMS chilled filter collects and holds more semivolatile mass than do the STN filters or that the STN filters lose some semivolatile material during sampling periods. Overall the agreement is quite reasonable and seems to indicate that one can reconstruct the FDMS mass measurement to within ~10% using the major species from the STN measurements at Queens. The comparison with the BAM is less encouraging. The reconstructed masses match well during the coldest periods, but diverge by more than 4 $\mu g/m^3$ in the middle of the year. This may be because of adsorption effects of water and other gases on the quartz filter during the summer months or some other systematic differences. The BAM results may be more correct than the FDMS, but we think it unlikely. Because the FRM and STN filter measurements are more closely matched by the FDMS measurements, we will concentrate on comparisons with the FDMS.

Another way to evaluate the methodology for mass reconstruction used here is to look at the correspondence of 24-hr averaged FDMS measured mass concentrations with reconstructed mass from the STN filters using the same assumptions as above. In essence this means OC is blank corrected and multiplied by 1.6, and water is estimated from thermodynamics. If this is done, daily comparisons can be illustrated as either scatter plots or as a time-series plot. Figure 7 shows the time-series plot for

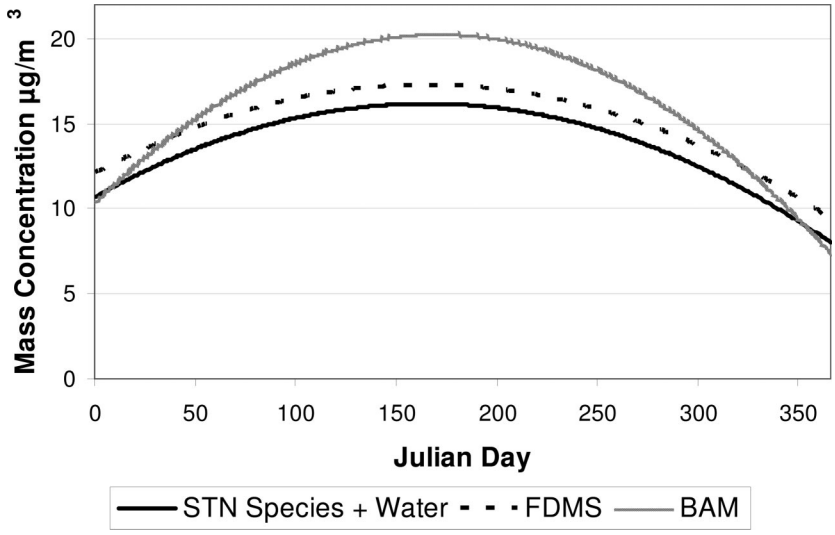


Figure 6. Reconstruction of species mass contributions at Queens using the polynomial fits for the individual species compared with the polynomial fit for the FDMS TEOM mass. Species included in the reconstruction are sulfate, nitrate, ammonium, EC, OC ×1.6, trace elements, and calculated water.

Queens in 2004. The scatter plot is highly correlated with an R^2 of 0.92.

A similar analysis was done using the Addison data for 2003 and 2004. As shown in Figure 8, the result for this site is not as encouraging. The consistent shape of the curves is absent in this case. Although it is difficult to be sure, we suspect that the FDMS data at the beginning and end of the year may be the problem. More specifically, the curvature in the fit for the FDMS data is ~60% greater than the curvature for the FRM data, and the FDMS fit yields mass concentrations that are unreasonably low at the beginning and end of the year where there is no valid FDMS data. Figure 8 begins at Julian day 53 because of the lack of valid FDMS data earlier than that for either 2003 or 2004. If one focuses only on days 180–280, the two curves look like the curves for Queens, only closer together. Clearly more work is needed here to develop a more reliable dataset to test the analytical method at this location.

Explanation of FDMS/FRM Difference at Queens

As noted above, there are reproducible differences between the FRM, FDMS, and standard TEOM measured values for $PM_{2.5}$ mass concentration. The standard TEOM reports the lowest concentrations, often called the nonvolatile or stable aerosol mass. The FRM captures a fraction, but not all, of the volatile components of the aerosol mass.[14] The FRM mass is greater than the standard TEOM mass, but less than the FDMS mass. The FDMS was designed to capture both the volatile and nonvolatile components of the aerosol mass. The differences between mass concentrations measured by these three methods are illustrated in Figure 9. The top curve is the second order polynomial fit to the daily differences between the 24-hr FDMS and FRM mass concentrations for Queens in 2004. This difference is near zero during the middle of the winter, and increases to 3.7 $\mu g/m^3$ in the summer. Not surprisingly from the correlations shown in Figures 2 and

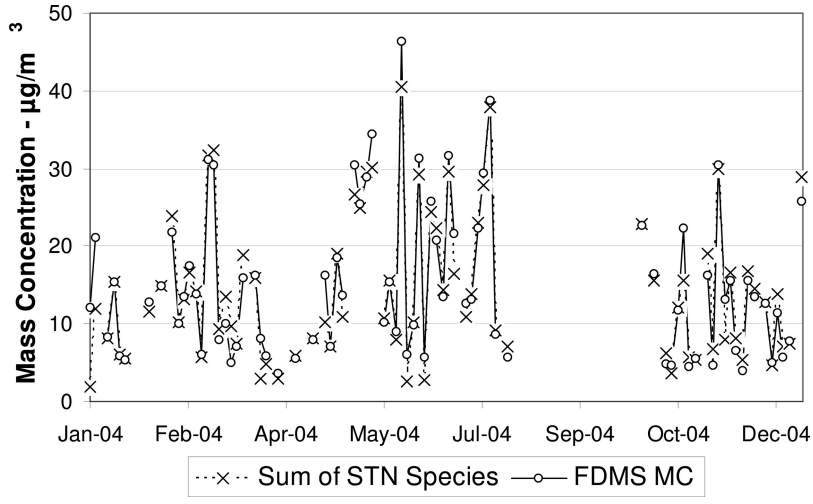


Figure 7. Time series of the reconstruction of species mass contributions at Queens using the 24-hr STN filter measurements for the individual species compared with the 24-hr averaged values for the FDMS TEOM mass. Species included in the reconstruction are as in Figure 6.

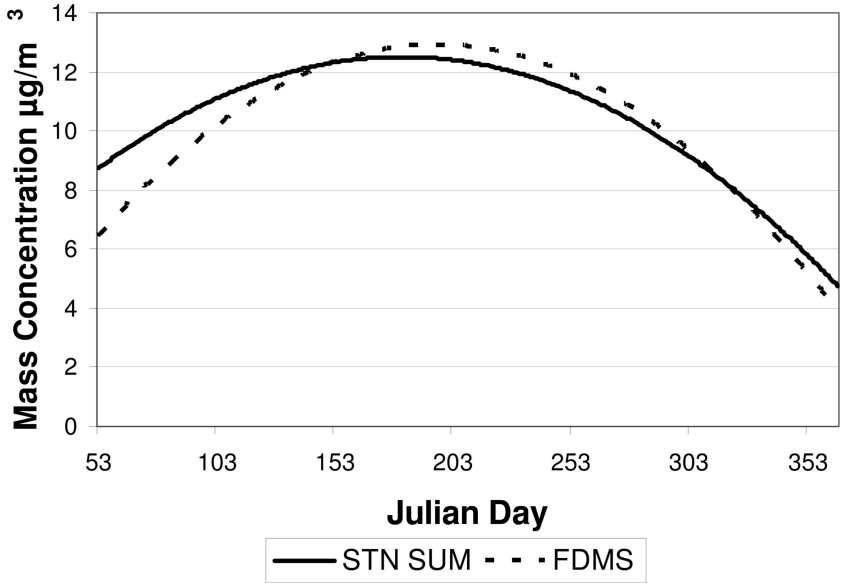


Figure 8. Reconstruction of species mass contributions at Addison using the polynomial fits for the individual species compared with the polynomial fit for the FDMS TEOM mass. Species included in the reconstruction are sulfate, nitrate, ammonium, EC, OC ×1.6, trace elements, and calculated water.

3, this difference is much smaller for the Addison site. The bottom curve is the fit to the daily differences between the standard TEOM and the FRM mass concentrations. The standard TEOM, which operates at 50 °C, clearly volatilizes a significant portion of the $PM_{2.5}$ mass in the winter, as has been shown many times.[3,7,15] In the summer, the standard TEOM approaches the FRM, but both of these methods capture substantially less mass than the FDMS.

Figure 9 gives a good representation of a number of measured quantities related to or sometimes referred to as fine particle mass. The zero line is the FRM mass concentration, the regulated quantity under the NAAQS. The bottom line is the stable or nonvolatile fine particle mass. This quantity can be routinely measured by a TEOM monitor operating at the default temperature of 50 °C. The area between the bottom curve and the zero line is the fraction of the semivolatile mass that is captured and retained by the FRM. At this site, most of the semivolatile fine particle mass is captured and retained during the winter, and most of it is lost during the summer. The top curve is the total fine particle mass, including the semivolatile material. The area between the top curve and the zero line is the semivolatile mass that is not captured and retained by the FRM. Model evaluation studies, as well as planning and regulatory applications, will benefit from an improved understanding of how these quantities relate to each other and, in particular, whether these results can support retrospective analyses of FRM measurements utilizing $PM_{2.5}$ chemical speciation data collected over the past five years.

Based on previous research,[2–4,14] we know that nitrate and OC are the two components that make up most, if not all, of the semivolatile fine particle mass lost from the FRM. Particle bound water, as estimated above, will be

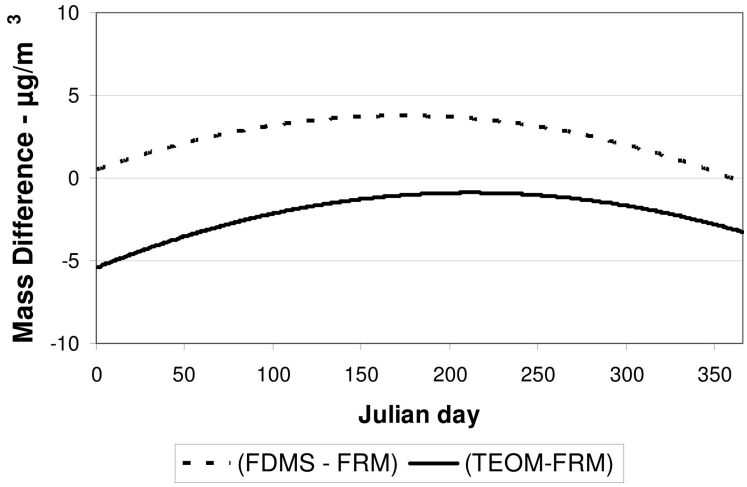


Figure 9. Julian day fits of the differences between the FDMS and the FRM and a collocated standard TEOM and the FRM for 2004 in Queens.

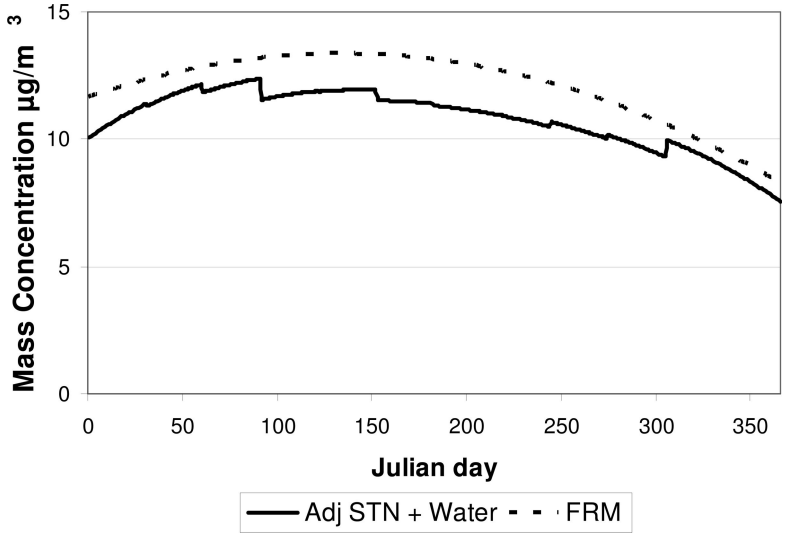


Figure 10. Reconstruction of FRM-equivalent mass compared with the polynomial fit for the 2004 FRM data at Queens. The same species are used in this reconstruction as were used in Figures 6 and 8, but the nitrate and OC are adjusted for losses from the FRM filter as described in the text.

retained by the FRM filters because the equilibration procedure for FRM and STN filters is identical. To estimate the fractional amount of nitrate lost from the FRM filters we use the nitrate retention study described by Frank.[14] In this study, the nitrate captured and retained on FRM Teflon sample filters was measured and compared with the nitrate captured and retained on the STN nylon filters at six sites in the Eastern United States. One of the sites was in the New York City borough of Bronx, 10 km from the Queens site. Monthly averages of the fraction of nitrate retained on the Teflon versus the nylon filters were measured, then estimated using an ammonium nitrate evaporation mode.[14,38–41] We use the Bronx monthly modeled nitrate retention results calculated by Frank.[14]

Estimating the amount of semivolatile OC lost from the FRM filter is an even more difficult task. Frank[14] discusses the many issues associated with determining the carbonaceous mass measured by the FRM. Given the many uncertainties associated with determining OC mass, and particularly the semivolatile fraction of the OC mass, Frank opts to estimate the total carbonaceous mass by mass balance (i.e., by subtracting sulfate, retained nitrate, ammonium, water, crustals, and other minor species from the measured FRM mass). Here, we attempt a less sophisticated estimate of the semivolatile OC retained on the FRM filter. We calculate the fraction of the semivolatile mass retained by the FRM using the polynomial fits for FDMS, FRM, and TEOM mass. This fraction, *f*, can be expressed as

$$f = \frac{(\mathrm{FRM} - \mathrm{TEOM})}{(\mathrm{FDMS} - \mathrm{TEOM})} \qquad (1)$$

The OC (blank corrected and multiplied by 1.6 as before) is multiplied by this fraction to estimate the amount of semivolatile OC retained on the FRM filter. With these adjustments for lost nitrate and OC, the reconstruction of the fitted species and estimated water, along with the fitted FRM curve, are shown in Figure 10. This adjusted reconstruction is systematically lower than the FRM by 0.5–1.8 $\mu g/m^3$. The discrepancy is larger in the summer months, possibly indicating that the rough method used for adjusting the OC is too aggressive during the warm months. Specifically, the fraction *f* in eq 1 will be too low and will over-adjust the OC if the FDMS measures significantly more OC (and possibly nitrate) than the STN sampler.

SUMMARY

As has been seen in previous work, measurements made with current generation continuous mass monitors are highly correlated with the FRM measurements. The FDMS TEOM monitor and the BAM measure 25% more mass than the FRM in Queens, averaged over 2004. For the rural Addison site, the FDMS TEOM monitor measures 9% more mass than the FRM over significant periods of 2003 and 2004. The linear correlation analysis does not illustrate the seasonal differences in these comparisons.

To capture the seasonal dependence of fine particle mass and its components, data from one or two complete years is used to calculate second order polynomial fits to various quantities, including FDMS, FRM, and standard TEOM mass concentrations and major chemical species from STN filter samples. The fits for the major species, along with an estimate for particle bound water, are used to create a reconstruction of $PM_{2.5}$ mass that is compared with the FDMS fit. The reconstruction is systematically lower than the FDMS by ~10% at Queens. Changing the estimated parameters, or detailed measurement issues may explain the difference at this site. The comparison of reconstructed mass and the FDMS mass is less consistent at the Addison site. Further interpretation using this analysis method for this rural site would require reliable FDMS data for the complete calendar year. Grover et al.[15] attempted to explain the difference between the FDMS and the 50 °C TEOM using measured nitrate and semivolatile organic material measurements for an eight-day period in July 2003 at Rubidoux, California. They found that with

few exceptions, the sum of ammonium nitrate and semivolatile organic material was equal to or slightly greater than the difference between the FDMS and heated TEOM monitors.

An attempt is made to reconstruct the FRM mass at Queens by using the same major species fits, but this time including an estimate of the amount of semivolatile nitrate and OC lost from the FRM filter (and retained by the STN filters). This reconstruction of the FRM mass is expected to be useful for states, tribes, and local agencies that are required to comply with the NAAQS. Again, the reconstruction at Queens is close, but systematically lower than the FRM. The greatest discrepancy is in the summer and may be because of an over-estimate of the volatile fraction of the OC mass by the FDMS.

The FDMS TEOM monitor has the attractive feature that the same instrument reports measurements of the nonvolatile and volatile mass concentrations. Although we have experienced numerous operational difficulties with the FDMS instruments, the added value of this partitioning of the total fine particle mass offers important insight into the character of fine particle pollution. This additional information will be very useful for model comparison and for developing pollution control strategies. The BAM also measures the sum of volatile and semivolatile $PM_{2.5}$ mass, making it suitable for reconstructions of total $PM_{2.5}$.

ACKNOWLEDGMENTS

This work was supported in part by the New York State Energy Research and Development Authority contract #4918ERTERES99, EPA cooperative agreement #R828060010; and New York State Department of Environmental Conservation contract #C004210. Although the research described in this article has been funded in part by EPA, it has not been subjected to the agency's required peer and policy review and therefore does not necessarily reflect the views of the agency and no official endorsement should be inferred. This work does not reflect the policies or practices of the New York State Department of Environmental Conservation. The authors thank John Spicer for instrument operation and filter collection at the Addison site, Mike Christopherson and Duane Villafana at the Queens site, and Neil Frank for helpful discussions.

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